

Building a Quant Risk Engine from Scratch — Part 2:

When Diversification Fails and Models Lie

Sobol QMC, Vine Copulas, Walk-Forward Backtesting,
and the Dollar Cost of Model Risk

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March 2026

Abstract

In Part 1, we built a stochastic volatility simulation engine and showed that Expected Shortfall estimates diverge by 30% across volatility models. Part 2 puts this engine to work. We demonstrate $191\times$ convergence improvement via Sobol quasi-Monte Carlo, show that equity tail dependence jumps from 0.01 to 0.49 during crises using vine copulas, validate on 1,053 days of real market data where Heston passes Basel traffic-light tests that GBM fails, and quantify that a Black–Scholes hedger underestimates tail losses by 6–12 $\times$ under realistic stochastic volatility. An interactive dashboard is available at <https://quantriskeng.streamlit.app>.

1. Faster Convergence — Quasi-Monte Carlo with Variance Reduction

Standard Monte Carlo converges at $O(N^{-1/2})$ —halving the error requires quadrupling the paths. For Expected Shortfall at the 99th percentile, where only 1% of simulated paths contribute to the estimate, this inefficiency is compounded: the effective sample size for the tail is $N/100$, and the resulting estimator has high variance. In production risk systems where ES must be computed across thousands of positions daily, convergence speed is not a theoretical nicety but a computational bottleneck.

Sobol sequences replace pseudorandom samples with low-discrepancy quasi-random numbers that fill the unit hypercube more uniformly. The theoretical convergence rate improves from the Monte Carlo $O(N^{-1/2})$ to:

$$\varepsilon_{\text{QMC}} \sim O\left(N^{-1}(\log N)^d\right) \quad (1)$$

where d is the effective dimension. For smooth integrands, this approaches $O(N^{-1})$ —achieving the same precision with the square root of the computational effort. We implement the Joe–Kuo direction numbers [7] with Brownian bridge path construction, which concentrates the most important dimensions (terminal value, early time steps) onto the lowest-discrepancy Sobol coordinates.

We supplement QMC with two classical variance reduction techniques. *Antithetic variates* exploit the symmetry of the standard normal distribution: for each path driven by innovations $Z_1, \dots, Z_n$, we simulate a mirror path using $-Z_1, \dots, -Z_n$, halving the variance of symmetric

functionals. *Control variates* use GBM—which has a known analytical solution—as a control for Heston:

$$\hat{\mu}_{CV} = \hat{\mu}_{\text{Heston}} + c^* (\mathbb{E}[g_{\text{GBM}}] - \hat{g}_{\text{GBM}}), \quad c^* = -\frac{\text{Cov}(\hat{Y}, \hat{G})}{\text{Var}(\hat{G})} \quad (2)$$

where both models are driven by identical random numbers. The achieved correlation of $\rho = 0.69$ reflects the fundamental difference between the models—as model risk increases, the GBM control becomes less effective. This is itself a diagnostic: the control variate correlation quantifies how far the stochastic volatility model has departed from the constant-volatility assumption.

Technique	Improvement
Sobol QMC (mean estimation)	$191\times$ lower standard error
Sobol QMC (ES estimation)	$\sim 1.5\times$ faster convergence
Control Variates (GBM $\rightarrow$ Heston)	28% SE reduction ($\rho = 0.69$)
Antithetic Variates	80% SE reduction

Table 1: Variance reduction results. Sobol QMC provides the largest gain for smooth statistics (mean, variance), while antithetic variates are most effective for tail risk metrics.

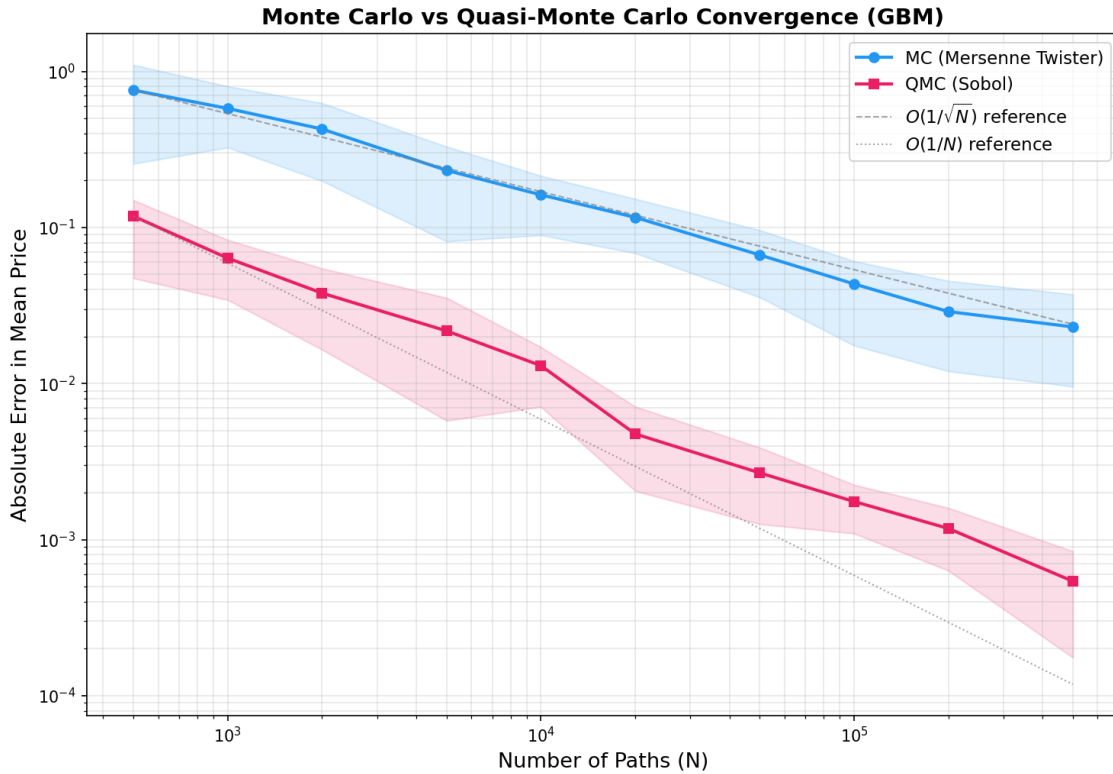


Figure 1: Convergence comparison: standard Monte Carlo (blue) vs. Sobol quasi-Monte Carlo (red). On log-log scale, MC follows the theoretical $O(N^{-1/2})$ slope while QMC achieves near- $O(N^{-1})$ convergence, requiring orders of magnitude fewer paths for equivalent precision.

2. When Diversification Fails — Vine Copulas and Tail Dependence

The most dangerous assumption in portfolio risk management is not about individual assets—it is about how they move together. In our empirical analysis (Part 1), we documented that equity pair correlations jump from 0.04–0.08 in calm markets to 0.37–0.42 during crashes. Pearson correlation cannot capture this asymmetry because it measures linear, symmetric dependence. A portfolio that appears well-diversified under normal conditions can become concentrated precisely when diversification matters most.

2.1 Copulas — Separating Margins from Dependence

Sklar’s theorem [12] provides the mathematical foundation for modeling dependence independently of marginal distributions:

$$F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d)) \quad (3)$$

The copula C captures the pure dependence structure, independent of how each individual asset’s returns are distributed. This separation is powerful: we can combine stochastic volatility marginals (from our C++ simulation engine) with realistic multivariate tail dependence (from the copula), without forcing either component to compromise.

2.2 Vine Copulas — Flexible High-Dimensional Dependence

For $d = 6$ assets (SPY, IBM, JPM, AAPL, TLT, GLD), specifying a full d -dimensional copula directly is impractical. Vine copulas [8] decompose the multivariate problem into a cascade of bivariate copulas arranged in a tree structure. Each bivariate pair is modeled with a Student- t copula that captures both upper and lower tail dependence, and the tree structure is selected automatically by maximizing Kendall’s τ across levels.

The key advantage over Gaussian copulas is the ability to model *asymmetric tail dependence*: the probability that two assets crash simultaneously can differ from the probability that they both surge. For equity portfolios, lower tail dependence systematically exceeds upper tail dependence—diversification erodes in crashes but not in rallies.

Finding	Value
Vine Copula ES ₉₉ (equal weight)	4.34%
Gaussian Copula ES ₉₉	3.78% (underestimates by 15%)
Independent ES ₉₉	2.62%
JPM risk contribution	38.1% of ES from 16.7% allocation
TLT risk contribution	−2.3% (crash hedge)
Crisis tail dependence (equity pairs)	λ_L jumps from 0.01 to 0.49

Table 2: Portfolio risk under vine copula dependence. JPM contributes $2.3\times$ its weight to tail risk; TLT provides genuine diversification even during crises.

The Euler allocation decomposes portfolio ES into additive per-asset contributions:

$$ES_i = \mathbb{E}[w_i r_i \mid r_{\text{portfolio}} \leq -\text{VaR}_\alpha], \quad \sum_{i=1}^d ES_i = ES_{\text{portfolio}} \quad (4)$$

The results are striking. Despite holding only one-sixth of the portfolio, JPM contributes 38.1% of tail risk because it is highly correlated with other equities in the left tail. SPY and AAPL contribute 21.2% and 22.6% respectively. GLD provides marginal diversification (2.3% contribution from 16.7% weight). TLT contributes *negatively* (-2.3%)—it is the only asset that genuinely hedges portfolio tail risk, because bonds tend to rally during equity crashes.

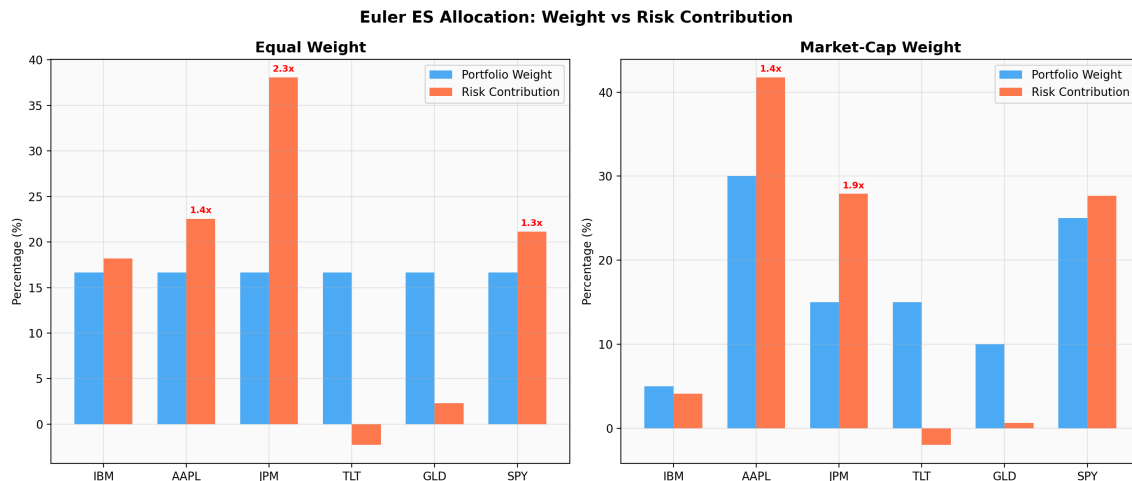


Figure 2: Euler risk allocation under vine copula. Left: portfolio weights (equal, 16.7% each). Right: ES₉₉ contributions. JPM dominates tail risk at 38.1%; TLT provides a negative contribution (diversification benefit that persists in crashes).

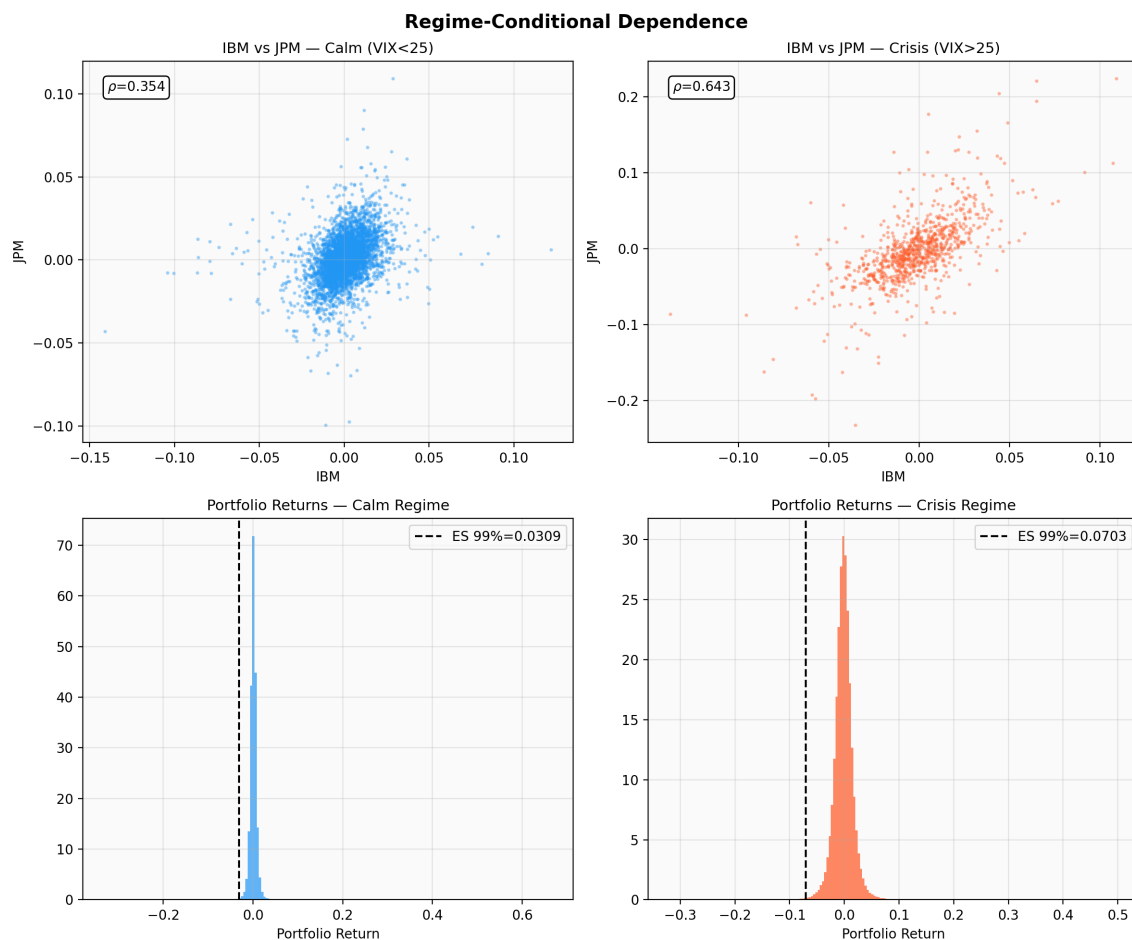


Figure 3: Tail dependence across market regimes. During calm periods (left), equity tail dependence λ_L is near zero (0.01–0.27). During crises (right), it jumps to 0.42–0.52 for equity pairs—a regime shift that Gaussian copulas cannot capture.

3. Proof on Real Data — Walk-Forward Backtesting

Theory and simulation are necessary but not sufficient. The critical question is whether our models produce calibrated risk estimates on *real market data*—data the models have never seen during calibration.

3.1 Setup

We run a 1,053-day walk-forward backtest (2022–2025) on SPY, IBM, and JPM. At each date t , we calibrate model parameters on all data prior to t , simulate 100,000 one-day-ahead scenarios, and compute VaR_{99} . We then compare with the realized return on day $t + 1$. Parameters are recalibrated via differential evolution every 60 trading days; the current variance v_0 is updated daily from a 20-day realized volatility window. This mimics how a production risk system would operate: periodic full recalibration with daily state updates.

3.2 Statistical Tests

The Kupiec test [9] evaluates *unconditional coverage*—whether the observed exceedance rate matches the target:

$$LR_{UC} = -2 \ln \frac{(1-p)^{T-x} p^x}{(1-\hat{p})^{T-x} \hat{p}^x} \sim \chi^2(1) \quad (5)$$

where p is the target exceedance rate, $\hat{p} = x/T$ is the observed rate, and x is the number of exceedances over T days.

The Christoffersen test [10] extends this to *conditional coverage*, testing whether exceedances are independent over time. It constructs the transition matrix of exceedance sequences and tests whether $P(\text{exceed}_{t+1} \mid \text{exceed}_t) = P(\text{exceed}_{t+1} \mid \text{no exceed}_t)$. Exceedance clustering indicates systematic model failure—the model is overconfident precisely during high-volatility regimes.

3.3 Results

Model	Exceedances	Rate	Kupiec p	Christ. p	Basel Zone
ARIMA	15	1.43%	0.193 (pass)	0.023 (fail)	GREEN
GBM	22	2.09%	0.002 (fail)	0.002 (fail)	YELLOW
Heston	14	1.33%	0.306 (pass)	0.237 (pass)	GREEN
Rough Heston	14	1.33%	0.306 (pass)	0.237 (pass)	GREEN

Table 3: Walk-forward VaR₉₉ backtest on SPY (1,053 trading days). Expected exceedances under correct calibration: ≈ 10.5 . GBM is formally rejected by Kupiec; ARIMA passes Kupiec but fails Christoffersen (clustered violations).

GBM produces 22 exceedances where approximately 10.5 are expected—formally rejected by Kupiec ($p = 0.002$). Its VaR is systematically too narrow because constant volatility cannot adapt to regime changes. ARIMA achieves acceptable unconditional coverage (15 exceedances, $p = 0.193$) but fails the Christoffersen independence test ($p = 0.023$) with a clustering ratio of $10.6\times$. Its violations come in bursts during volatile periods—arguably more dangerous than a uniform over-count, because clustered violations can compound into catastrophic multi-day losses.

Heston and Rough Heston both achieve Basel GREEN with 14 exceedances (1.33% vs. 1% target) and pass both Kupiec and Christoffersen tests. They succeed because the stochastic variance process adapts to regime changes: when realized volatility rises, v_0 is updated upward, widening the one-day VaR. The models are not perfect—14 exceedances is still above the expected 10.5—but they are statistically indistinguishable from correct calibration.

3.4 Regime Analysis

The regime decomposition reveals where models fail most dramatically:

Model	Calm Rate	Crisis Rate	Crisis Days	Target
ARIMA	0.00%	9.32%	161	1%
GBM	0.22%	12.42%	161	1%
Heston	0.45%	6.21%	161	1%
Rough Heston	0.45%	6.21%	161	1%

Table 4: VaR₉₉ exceedance rates by market regime. All models struggle in crisis periods (target: 1%), but Heston halves GBM’s crisis exceedance rate from 12.4% to 6.2%.

GBM’s crisis exceedance rate of 12.4% means that on roughly one in eight crisis days, losses exceed the stated VaR—over twelve times the target rate. ARIMA is even more extreme in its failure mode: zero exceedances during calm periods (overfitting) followed by 9.3% during crises. Heston reduces the crisis rate to 6.2%—still elevated, but halved relative to GBM. The improvement comes directly from the variance process adapting to regime changes.

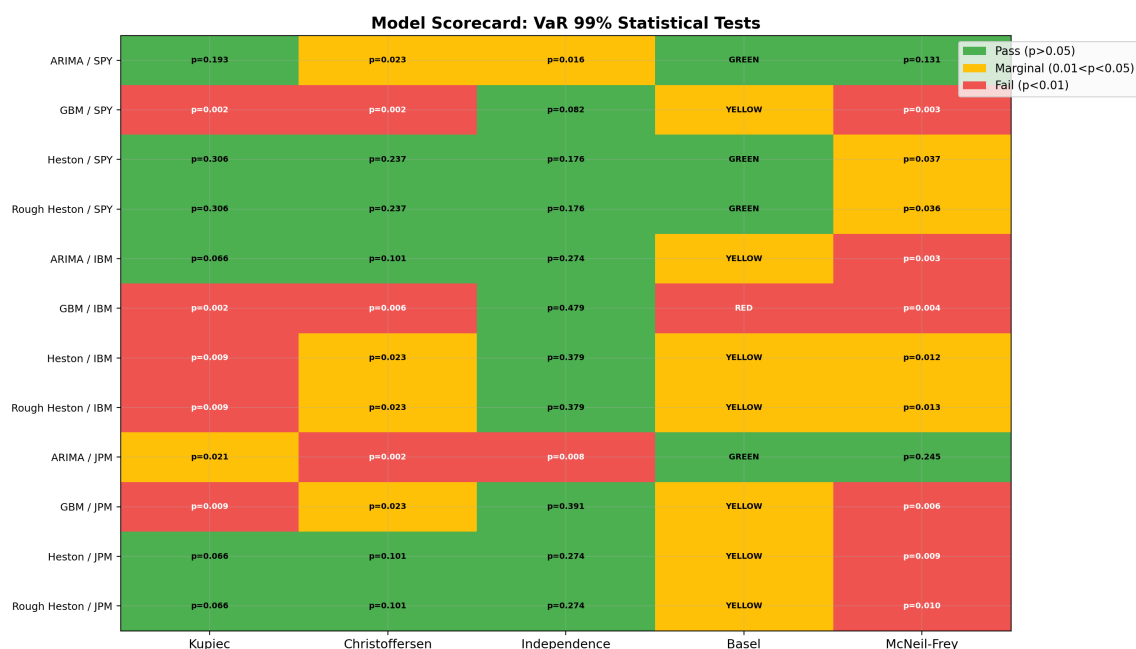


Figure 4: Model scorecard heatmap. Green cells indicate statistical test passage ($p > 0.05$); red cells indicate failure. Only Heston and Rough Heston achieve green across all tests at the 99% confidence level.

4. The Dollar Cost of Model Risk — Dynamic Hedging

The preceding sections quantify model risk in statistical terms—exceedance rates, p -values, percentage differences in ES. This section translates those abstractions into a concrete financial quantity: the profit-and-loss distribution of a hedged options position.

4.1 Setup

A trader sells a 3-month ATM call option on SPY ($S_0 = 100$, $K = 100$) and collects a Black–Scholes premium of \$4.62. The trader delta-hedges daily using the Black–Scholes formula with

assumed constant volatility $\sigma = 0.20$:

$$\Delta_{\text{BS}} = N(d_1), \quad d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)\tau}{\sigma\sqrt{\tau}} \quad (6)$$

Under GBM (when the model is correct), the hedge is nearly perfect—the trader earns approximately the risk-free rate on the premium collected. Under stochastic volatility (when the model is *wrong*), systematic hedge errors accumulate because the true delta depends on the instantaneous variance v_t , which the trader ignores.

4.2 Results

We simulate 100,000 hedge P&L paths under three “true” market dynamics using calibrated SPY parameters:

True Dynamics	Hedge Strategy	P&L Std	ES ₉₉	ES Ratio	$P(\text{loss} > 2\times)$
GBM	BS constant σ	\$0.52	\$0.90	1.0×	0.00%
Heston	BS constant σ	\$1.72	\$5.53	6.1×	0.06%
Heston	BS local vol	\$1.74	\$4.15	4.6×	0.00%
Rough Heston	BS constant σ	\$2.34	\$10.42	11.6×	0.55%
Rough Heston	BS local vol	\$2.29	\$7.63	8.5×	0.11%

Table 5: Hedge P&L statistics per \$100 notional. “ES Ratio” is relative to the GBM baseline. Under Rough Heston, 0.55% of paths produce losses exceeding twice the collected premium—a scenario the Black–Scholes framework considers impossible.

Under GBM, the hedge works as advertised: ES₉₉ is \$0.90, well within the \$4.62 premium collected. Under Heston, ES₉₉ jumps to \$5.53—a 6× increase that exceeds the premium. Under Rough Heston, where volatility is rough and harder to track, ES₉₉ reaches \$10.42—a 12× increase. The trader who assumes Black–Scholes is correct would set aside reserves for \$0.90 of tail risk; the actual tail risk is \$10.42.

On a \$10M notional position, this translates to a hidden tail risk of \$553,000–\$1,042,000 that the Black–Scholes-assuming trader does not see. This is not a hypothetical stress scenario—it is the expected 99th-percentile loss under empirically calibrated market dynamics.

Even a simple improvement—using $\sigma_t = \sqrt{v_t}$ (local volatility updated daily from recent realized variance) instead of constant σ —reduces Heston ES₉₉ from \$5.53 to \$4.15, a 25% improvement. Under Rough Heston, local vol hedging reduces ES₉₉ from \$10.42 to \$7.63, a 27% reduction. But the residual risk remains substantial: model risk cannot be hedged away entirely; it can only be acknowledged and managed.

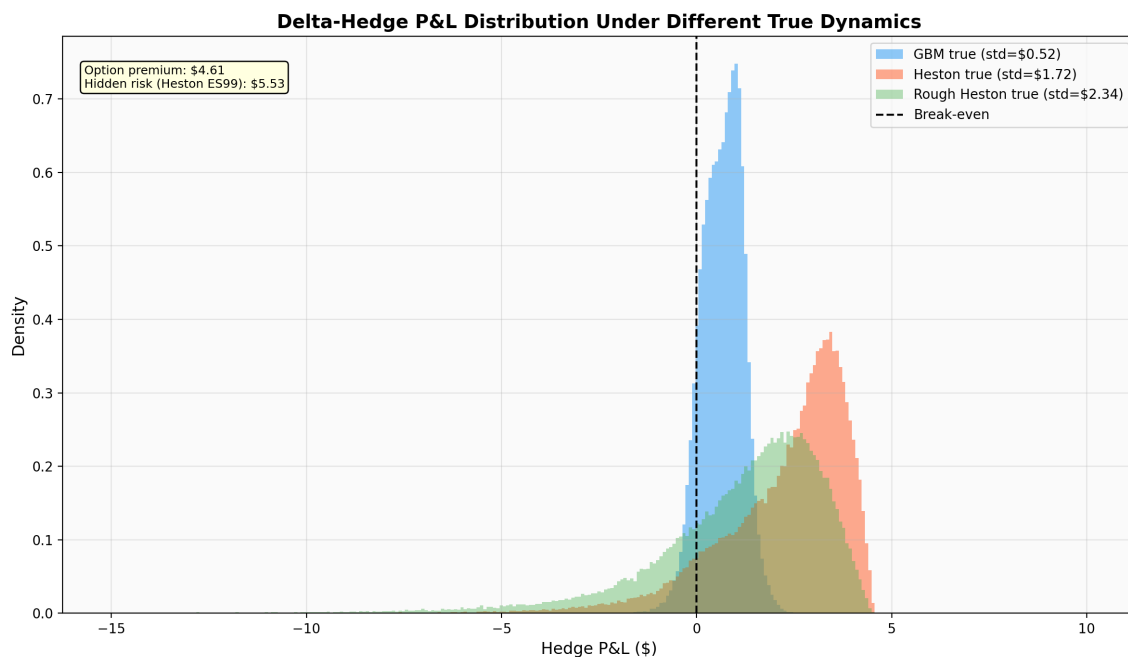


Figure 5: Hedge P&L distributions under three market dynamics. The Black–Scholes-assuming trader (blue) sees a tight distribution around zero. Under Heston (orange) and Rough Heston (red), the true P&L develops heavy left tails that the trader does not anticipate.

5. Interactive Dashboard

To make these results accessible beyond static figures, we built an interactive Streamlit dashboard deployed at <https://quantriskeng.streamlit.app>. The application provides five pages: a Model Explorer for real-time parameter adjustment with immediate visual feedback, a three-model risk comparison, portfolio risk analysis with copula-dependent tail scenarios, walk-forward backtest visualization with statistical test details, and dynamic hedging analysis with position scaling.

The dashboard uses a dual-mode architecture: locally, it connects to the C++/pybind11 simulation engine for full-speed Monte Carlo with up to 500,000 paths; on cloud deployment, it falls back to a pure Python simulation backend that supports any parameter combination interactively. All five pages render correctly in both modes, allowing reviewers and collaborators to explore the full risk engine without compiling C++ code.

6. What We Learned

Three findings stand out from this project, each with direct implications for how risk is managed in practice.

Model risk is quantifiable and material. The choice of volatility model changes ES_{99} by 30%. The choice of dependence model changes portfolio ES by 15%. Combined, using GBM with a Gaussian copula instead of Heston with a vine copula underestimates tail risk by over 40%. These are not edge cases or stress scenarios—they are the baseline estimates under empirically calibrated parameters.

Prediction intervals, not point forecasts. No model—ARIMA, Random Forest, XGBoost, or any stochastic volatility model—can reliably predict tomorrow’s return. But stochastic volatility models can produce calibrated prediction intervals that contain the true value at the stated confidence level. Our backtest confirms this: Heston passes both Kupiec and Christoffersen tests on 1,053 days of real SPY data, achieving Basel GREEN where GBM fails. Risk management does not need accurate point forecasts; it needs honest uncertainty quantification.

Model risk has direct financial consequences. A trader delta-hedging with Black–Scholes faces 6–12× more tail risk than the model predicts. On a \$10M position, this is a hidden exposure of \$553K–\$1.04M. This is not an academic curiosity; it is a P&L line item that materializes during crises—precisely when institutions can least afford unexpected losses.

What’s next. CUDA GPU acceleration for nested Monte Carlo—computing the distribution of ES itself, quantifying estimation uncertainty as meta-risk. Calibration on implied volatility surfaces for production-quality parameter estimation. The code is open-source at <https://github.com/notgabrielpop/quant-risk-engine>, and the interactive dashboard is live at <https://quantriskeng.streamlit.app>.

The question was never “What will the price be?” The question is “What does the full landscape of possibilities look like, how confident are we in our map, and what does it cost when the map is wrong?”

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